COMP8240 Project Final Report

Joseph Belbin
Macquarie University
joseph.belbin@students.mq.edu.au
https://github.com/josephbelbin-mq/iTransformer
https://github.com/josephbelbin-mq/PatchTST

Abstract

This project reproduces and extends the evaluation of iTransformer, a transformer architecture for multivariate time-series forecasting. The original results reported by Liu et al. were successfully replicated on four benchmark datasets, with reproduced forecasting errors typically differing by less than 5% from the published values. Additional experiments were conducted on two real-world datasets and a controlled synthetic benchmark. Results showed that iTransformer did not consistently outperform simpler forecasting approaches on the real-world datasets, while a clear advantage emerged under synthetic conditions characterised by strong nonlinear cross-variate interactions. These findings suggest that the benefits of iTransformer are most likely to emerge when substantial nonlinear relationships exist between variates.

1 Introduction

Replication and reproduction are fundamental components of the scientific method. While definitions differ between fields and research communities, for the purposes of this study, replication involves repeating an experiment using the same experimental setup to verify that consistent results can be generated, whereas reproduction applies the same methodology in a different setting, to ensure that results are generalisable. Many scientific fields are currently experiencing a *reproducibility crisis* (Baker, 2016), whereby the findings of landmark studies cannot be reliably reproduced or replicated. This crisis also affects the field of Computer Science (Cockburn et al., 2020).

This project seeks to assess the extent to which a recently published and influential computer science paper can be replicated and reproduced.

2 Description of the Source Paper

This project reproduces the paper *iTransformer: Inverted Transformers Are Effective for Time Series Forecasting* (Liu et al., 2024).Liu et al. investigate the use of Transformers for multivariate time series forecasting, a task in which future observations must be predicted from historical measurements of multiple, related variables.

At the time of the iTransformer paper, Transformer-based forecasting models had attracted considerable attention (Zhou et al., 2021; Lai et al., 2018; Wu et al., 2021; Liu et al., 2022), however they were often outperformed by simple linear models in terms of both forecasting accuracy and computational efficiency. The authors argue that this is largely because conventional Transformers use self-attention to model relationships between timesteps, whereas many forecasting problems depend more heavily on relationships between variates. In a conventional Transformer, each timestep is represented as a token, consisting of the values of all variates at that point in time. Consequently, self-attention learns relationships between different points in time.

To address this, Liu et al. propose an inverted input representation for Transformer-based forecasting models, as shown in Figure 1. Rather than having a token for each timestep, each variate is represented as a token containing its historical observations. As a result, the input tensor is effectively transposed, so that each token corresponds to a complete history of a single variate rather than a snapshot of all variates at a single timestep. This inversion allows self-attention to learn re-

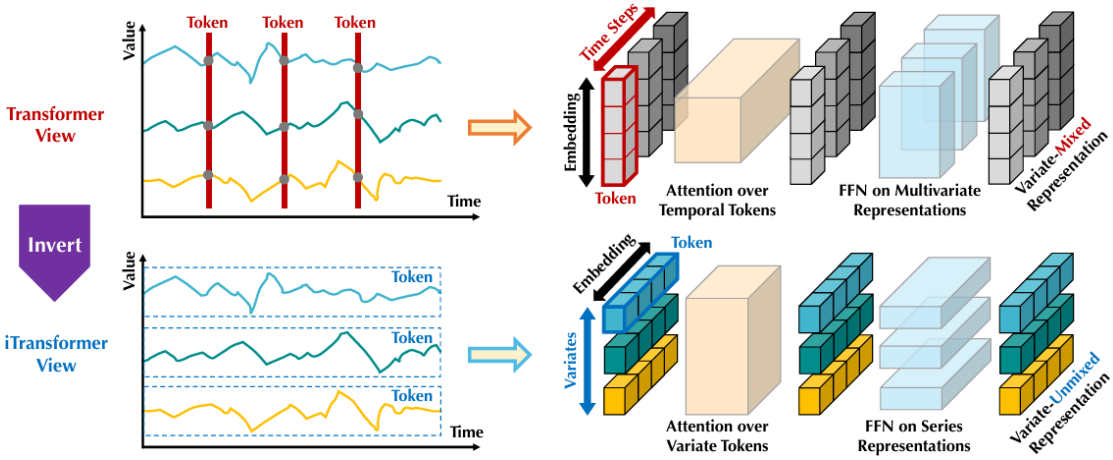


Figure 1: The iTransformer architecture. Reproduced from (Liu et al., 2024).

lationships between variates, while temporal patterns are primarily captured by the feed-forward network. Importantly, this functionality is achieved without modifying the underlying Transformer architecture, instead relying solely on a different representation of the input data.

The authors evaluate iTransformer using Mean Squared Error (MSE) and Mean Absolute Error (MAE), comparing its performance against a range of existing forecasting approaches. iTransformer achieves strong forecasting performance across multiple benchmark datasets and forecasting horizons, outperforming many established forecasting models.

3 Justification for Paper Choice

The iTransformer paper was selected for its impact, publication venue and suitability for reproduction. The paper was originally presented at the 2024 International Conference on Learning Representations (ICLR), where it was accepted as a *Spotlight* paper. ICLR is a top tier conference, having a CORE ranking of A*, and is ranked 2nd in the Artificial Intelligence subcategory by Google Scholar Metrics.

At the time of writing, the paper had accumulated more than 4,400 citations, demonstrating substantial interest from the research community in a relatively short period of time. This level of engagement suggests that the proposed method has been influential in the development of time-series forecasting research.

From the perspective of replication and reproduction, the paper is particularly suitable. All datasets in the paper are publicly available, and an official iTransformer implementation with source code is available on GitHub at thuml/iTransformer. Furthermore, preliminary experiments confirmed that this implementation could be executed successfully on the computational resources available to this project. These factors make the paper well suited to the objectives of this study.

4 Description of the Original Datasets

The iTransformer paper evaluates forecasting performance using a set of seven real-world datasets, which originate from a mixture of public data sources and benchmark preprocessing pipelines introduced in previous forecasting studies. These datasets, summarised in Table 1, span domains including energy consumption, transportation, weather prediction and power systems, allowing evaluation across a range of forecasting settings.

Dataset	# Variates	Domain
ECL	321	Electricity Load
ETT	7	Power Systems
Weather	21	Meteorology
Exchange	8	Finance
Traffic	862	Transportation
Solar-Energy	137	Solar Energy
PEMS	170–883	Transportation

Table 1: Datasets used in the iTransformer paper

Each dataset is provided as a CSV file in which rows correspond to timestamps and columns correspond to variates. Consequently, each dataset may be represented as a multivariate time series, where multiple related variables are observed simultaneously over time. The official implementation loads each dataset and partitions it into training, validation and test segments. Input-target pairs are then generated using a sliding window approach, where a fixed lookback window of historical observations is used to predict observations over a future forecasting horizon.

Figure 2 illustrates a representative subset of the Exchange dataset used in the original iTransformer evaluation. The dataset consists of eight daily exchange-rate series and is used as a multivariate forecasting benchmark. Similar multivariate time-series structures are used throughout the benchmark suite, although the number of variates, sampling frequencies and application domains differ between datasets.

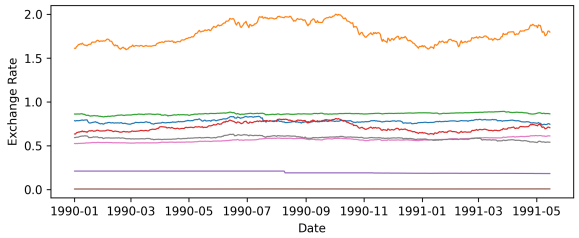


Figure 2: Representative segment of Exchange Time Series Data

For the present study, four datasets were selected for reproduction: ECL, Traffic, Weather and Exchange. These datasets span a range of forecasting applications and demonstrate substantial variation in both the number of variates and dataset size.

5 Construction of New Data

To complement the benchmark datasets used by Liu et al., a synthetic multivariate time-series dataset was generated. The aim of this dataset was to provide a controlled environment in which the strength of inter-variable interactions and the degree of nonlinearity could be systematically varied. This allowed the behaviour of different forecasting models to be evaluated under known data-generating processes. The model was based on a multivariate Vector Autoregressive (VAR) process. Full implementation details, including coefficient generation, stability constraints, and noise parameters, are available in the accompanying source code repository.

A conventional VAR process models each variable as a linear combination of previous observations and a random noise term

$$\mathbf{x}_t = \sum_{d \in D} \mathbf{A}_d \mathbf{x}_{t-d} + \boldsymbol{\epsilon}_t$$

where D is the set of memory lags, $\mathbf{A}_d$ contains the autoregressive coefficients for lag d and $\boldsymbol{\epsilon}_t$ is a random noise vector. The noise term was sampled independently from a zero-mean Gaussian distribution with standard deviation $\sigma = 0.1$.

For a system containing N variates, each coefficient matrix $\mathbf{A}_d$ has dimensions $N \times N$. For example, in a dataset containing eight variates, each $\mathbf{A}_d$ would be an 8×8 matrix.

To independently control temporal memory and interactions between variables, the coefficient matrices were decomposed into diagonal and off-diagonal components. Diagonal entries represent the influence of a variable's own history, while off-diagonal entries represent interactions between different variables.

$$\mathbf{x}_t = \sum_{d \in D} \left(\mathbf{A}_d^{\text{diag}} \mathbf{x}_{t-d} + \lambda \mathbf{B}_d \mathbf{x}_{t-d} \right) + \boldsymbol{\epsilon}_t$$

where $\mathbf{A}_d^{\text{diag}}$ contains only the diagonal autoregressive coefficients, $\mathbf{B}_d$ contains the off-diagonal coefficients, and λ acts as a scaling parameter that modulates the contribution of cross-variate interactions to the generated time series.

To introduce nonlinear dependencies between variables, the interaction term was extended to include both linear and nonlinear components:

$$\mathbf{I}_{t,d} = (1 - \gamma) \mathbf{B}_d \mathbf{x}_{t-d} + \gamma f(\mathbf{B}_d \mathbf{x}_{t-d})$$

$$\mathbf{x}_t = \sum_{d \in D} \left(\mathbf{A}_d^{\text{diag}} \mathbf{x}_{t-d} + \lambda \mathbf{I}_{t,d} \right) + \boldsymbol{\epsilon}_t$$

where $f(\cdot)$ is a nonlinear transformation based on the square-root function and $\gamma \in$

$[0, 1]$ controls the relative contribution of the linear and nonlinear interaction terms. Special cases arise when γ is set to 0 or 1. When $\gamma = 0$, the process reduces to a purely linear VAR model. When $\gamma = 1$, all cross-variate interactions are nonlinear. Intermediate values of γ produce varying degrees of nonlinearity.

Several nonlinear transformations, including the sign function ($\text{sign}(x)$) and logarithmic functions, were explored during dataset development. The square-root transformation was ultimately selected for the experiments reported here because it introduced substantial nonlinearity while maintaining stable time-series dynamics.

Memory dependencies were introduced at lags of 1, 2, 4, 8, 16, 32, 64 and 128 timesteps, allowing the generated processes to exhibit both short-term and long-term temporal dependencies. Exponentially spaced lags were selected to capture interactions across multiple temporal scales while avoiding the need to estimate a separate coefficient matrix for every preceding timestep, which would substantially increase model complexity and produce an unrealistically dense dependency structure.

Each synthetic dataset consisted of M independent simulated trajectories. Each trajectory represented an independent realization of the underlying data-generating process, analogous to a separate observation period within a larger dataset. For each trajectory, a multivariate time series consisting of N variates was generated. To reduce the influence of arbitrary initial conditions, a burn-in period was applied and the corresponding observations were discarded prior to constructing each trajectory. This allowed the process to reach a stable operating regime before samples were collected. Coefficient matrices were generated subject to stability constraints to ensure that the resulting process remained bounded and did not exhibit explosive behaviour.

For storage efficiency, these M trajectories were stored within a single dataset. Forecasting windows for training and evaluation were constrained to remain within individual trajectories, ensuring that no samples crossed simulation boundaries.

As different values of λ and γ alter the statistical properties of the generated time series, including their variance and amplitude, absolute MSE values are not directly comparable across synthetic datasets constructed using different parameter settings. Consequently, comparisons focus on the relative performance of competing models within a given dataset configuration.

5.1 Persistent State Dataset

While the synthetic VAR dataset allows the strength of temporal dependencies and nonlinear interactions to be controlled directly, many real-world time series also exhibit periods of changing behaviour driven by unobserved factores. To investigate this scenario, an additional persistence mechanism was introduced for a subset of experiments.

A hidden binary state variable $s_t \in \{0, 1\}$ was added to the data-generating process. At each timestep, the state remained unchanged with probability p, and switched with probability $1 - p$. When $s_t = 0$, the series evolved according to the diagonal autoregressive component only, while $s_t = 1$ activated the full interaction model. The hidden state was not observed by the forecasting model. These experiments were conducted only for the $\lambda = 1$, $\gamma = 1$ configuration, allowing the effect of persistent latent structure to be evaluated under a strongly nonlinear setting.

$$\mathbf{x}_t = \begin{cases} \sum_{d \in D} \mathbf{A}_d^{\text{diag}} \mathbf{x}_{t-d} + \boldsymbol{\epsilon}_t, & s_t = 0 \\ \sum_{d \in D} \left(\mathbf{A}_d^{\text{diag}} \mathbf{x}_{t-d} + \lambda I(\mathbf{x}_{t-d}) \right) + \boldsymbol{\epsilon}_t, & s_t = 1 \end{cases}$$

6 Additional Real-World Datasets

6.1 Beijing PM2.5

Source: Monash, UEA & UCR Beijing PM2.5 Dataset

The Beijing PM2.5 dataset (Tan et al., 2020) contains air quality and meteorological measurements taken in Beijing, China, between 2013 and 2017. The dataset includes hourly observations of nine environmental variables: hourly air pollutants (SO2, NO2, CO and O3), temperature, pressure, dew point, rainfall and windspeed. The original forecasting task for this dataset was regression of these hourly predictors to predict a single, daily PM2.5 concentration value. The original

dataset was published with predefined training and test sets. As iTransformer performs its own chronological training, validation and test split, only the training portion of the original dataset was used.

The dataset contained observations from twelve monitoring stations. Each station record was validated using the supplied timestamps and analysed as an independent time series. This analysis revealed a small number of multi-day gaps within individual station records. To retain information across all monitoring stations, station-level series were aligned using their timestamps and combined into a single multivariate panel. Where time periods were missing observations caused by short station outages (less than 24 hours), measurements were imputed using forward and backward filling. Longer missing periods were not imputed and instead defined segment boundaries. The aligned panel was subsequently partitioned into continuous multistation segments, ensuring that forecasting windows never crossed gaps in the original data.

As the PM2.5 target was supplied as a daily regression label rather than a continuous hourly series, it could not be naturally incorporated into the forecasting framework. Consequently, the forecasting experiments were performed on the observed environmental variables only.

The final dataset consisted of eleven continuous multistation segments containing 108 variates (12 monitoring stations $\times$ 9 observed variables). Segment lengths ranged from 936 to 2,181 hourly observations, with a median length of 1,253 hours.

6.2 Space Weather Indices

Source: Hugging Face Space Weather Indices Dataset

The Space Weather Indices dataset (Simon, 2026) contains daily geomagnetic and solar activity measurements recorded since 1957. It includes Kp and Ap geomagnetic indices, F10.7 solar radio flux measurements, and international sunspot numbers. These variables are commonly used to characterise solar activity and its interaction with the Earth's magnetosphere.

An exploratory data analysis identified a potential data-quality issue involving the alignment of several derived variables, which was subsequently reported to the dataset maintainer. The affected derived variables were excluded from all forecasting experiments. The dataset was further filtered to retain only observed measurements, and metadata fields relating to observation type and storm classification were removed. Several derived variables, including aggregated geomagnetic indices and smoothed F10.7 solar flux measurements, were excluded to reduce redundancy and limit the influence of variables that could be computed directly from other observations. The remaining observations were sorted chronologically and exported in the CSV format expected by the iTransformer data loader. The resulting dataset contained approximately 25,000 daily observations across 23 variates.

7 Experimental Setup

7.1 Models

Four forecasting models were evaluated: iTransformer (Liu et al., 2024), PatchTST (Nie et al., 2023), DLinear (Zeng et al., 2022), and a custom Linear baseline. The Linear baseline was implemented specifically for this project to provide a simple reference model and sanity check for the transformer-based forecasting approaches. Experiments were based on the official implementations released by the respective authors. PatchTST and DLinear were evaluated using the implementation provided in the official PatchTST repository. Minor modifications were made to improve compatibility with the local software and hardware environment, and to support additional logging and result-management functionality. No changes were made to the underlying model architectures or forecasting methodologies.

7.2 Experimental Environment

Experiments were conducted on a laptop equipped with an NVIDIA GeForce RTX 3070 GPU with 8 GB of memory. All experiments were run under Ubuntu 24.04 using Windows Subsystem for Linux (WSL) and a Python virtual environment. Model training was performed using PyTorch 2.0.0 with CUDA 11.7.

Table 2: Comparison of published and reproduced MSE results across benchmark datasets. Lower values indicate better forecasting performance.

	Published			Reproduced		
Dataset	iTransformer	PatchTST	DLinear	iTransformer	PatchTST	DLinear
Traffic	0.395	0.462	0.650	0.393	0.478	0.650
Exchange	0.086	0.088	0.088	0.088	0.083	0.088
Weather	0.174	0.177	0.196	0.174	0.172	0.196
ECL	0.148	0.181	0.197	0.148	0.165	0.195

7.3 Hyperparameters

For consistency, all primary experiments used a lookback window of 96 timesteps and a forecasting horizon of 96 timesteps. This ensured a fair comparison between models while remaining feasible within the available computational resources. Larger lookback windows substantially increased training time and often exceeded the available GPU memory. Additionally, the maximum number of training epochs were fixed at 10 and the early stopping patience was fixed at 3 for all models.

Unless otherwise stated, experiments followed the recommended configurations provided in the official implementations released by the respective authors. Minor hyperparameter adjustments were made where required to support new datasets and available computational resources.

7.4 Evaluation metrics

Following Liu et al. (Liu et al., 2024), forecasting performance was evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE). Lower values indicate improved forecasting accuracy. MSE was used as the primary evaluation metric, while MAE was also reported to provide a complementary measure of forecasting error that is less sensitive to large outliers.

8 Results

8.1 Reproduction of published results

Table 2 compares the forecasting performance of the reproduced implementation against the results reported by Liu et al. across several benchmark test sets.

Overall, the reproduced results were broadly consistent with those reported in the original paper. Although minor changes in model ranking were observed for some datasets, reproduced forecasting errors remained close to the published values and the overall conclusions of the original paper were preserved. These results provide evidence that the principal findings of Liu et al. can be reproduced using the published implementations and the computational resources available to this project.

8.2 Performance on Additional Real-World Datasets

8.2.1 Beijing PM2.5

Table 3 summarises the forecasting performance of the evaluated models on the reconstructed Beijing PM2.5 dataset.

Table 3: Forecasting performance on the reconstructed Beijing PM2.5 multi-station dataset.

Model	MSE	MAE
Linear	0.680	0.472
DLinear	0.684	0.473
PatchTST	0.698	0.457
iTransformer	0.714	0.473

The simple Linear model achieved the lowest MSE, while PatchTST achieved the lowest MAE. iTransformer achieved competitive performance but did not achieve the best result under either evaluation metric.

8.2.2 Space Weather

Table 4 summarises the forecasting performance of the evaluated models on the reconstructed Space Weather dataset.

PatchTST achieved the lowest MSE and MAE. The remaining models produced broadly comparable forecasting performance, with relatively small differences between their forecasting errors.

Table 4: Forecasting performance on the Space Weather dataset.

Model	MSE	MAE
Linear	0.551	0.523
DLinear	0.552	0.552
PatchTST	0.517	0.461
iTransformer	0.573	0.504

8.2.3 Variate analysis

To better understand these results, a simple autoregressive linear regression model was independently fit to each variate for both data sets. Each model used the previous 96 observations to predict the next value, using the same lookback window as the primary forecasting experiments. The coefficient of determination (R^2) was then used as an indicator of linear predictability.

Most variables in the reconstructed Beijing PM2.5 multi-station dataset exhibited high linear predictability. Across the 108 observed variables, the median coefficient of determination achieved by a 96-lag autoregressive linear model was 0.919, with many variables achieving values in excess of 0.99. These results indicate substantial persistence and linear structure within the observed environmental variables.

In contrast, the Space Weather dataset showed a wider range of linear predictability. Although several variables, including the F10.7 solar radio flux and sunspot number measurements, exhibited high linear predictability ($R^2 \approx 0.95$), many geomagnetic activity indices exhibited substantially lower linear predictability ($R^2 < 0.4$). Despite the presence of these less predictable variables, no clear and consistent advantage was observed for the transformer-based models under the evaluated configurations.

It should also be noted that both real-world datasets exhibited forms of non-homogeneity that are commonly encountered in operational forecasting settings. The Beijing dataset contained station outages and discontinuities that required segmentation during reconstruction, while the Space Weather dataset exhibited noticeable distributional differences between the training, validation, and test partitions. These characteristics may have increased forecasting difficulty for all evaluated models and complicate any explanation based solely on variable-level linearity.

Detailed per-variable linearity results are provided in Appendix A.

8.3 Performance on Synthetic Datasets

8.3.1 Effects of Interaction Strength and Non-Linearity

To investigate the effects of interaction strength and nonlinearity on forecasting performance, models were evaluated on synthetic datasets generated using the procedure described in Section 5. Table 5 reports representative results for three key parameter configurations. Absolute MSE values should not be compared between rows, as changes in λ and γ alter the variance and amplitude of the generated process. Instead, emphasis should be placed on the relative ranking of the models within each configuration. Figure 3 summarises the winning model across the full λ and γ parameter sweep. Cell colour indicates the winning model, while colour intensity indicates the margin of victory relative to the second-best model.

Table 5: Forecasting performance on representative synthetic datasets.

λ	γ	Linear	DLinear	PatchTST	iTrans.
0	0	0.996	0.996	1.008	1.007
1	0	0.991	0.991	1.004	0.978
1	1	0.347	0.348	0.355	0.272

When interactions were disabled ($\lambda = 0$), all models achieved almost identical forecasting performance. Introducing linear interactions ($\lambda = 1$, $\gamma = 0$) resulted in only minor differences between models, although iTransformer achieved the lowest MSE. In contrast, when nonlinear interactions were introduced ($\lambda = 1$, $\gamma = 1$), iTransformer substantially outperformed all baseline models. Full MSE and MAE results for the complete λ and γ parameter sweep are provided in Appendix B.

An unexpected result of the synthetic experiments was the relatively poor performance of PatchTST. Additional sensitivity tests were therefore conducted using a longer lookback window. The results are shown in Table 6.

Increasing the lookback window from 96 to

Table 6: Effect of lookback window on forecasting performance for the fully nonlinear synthetic dataset ($\lambda = 1$, $\gamma = 1$).

	PatchTST	PatchTST	iTrans.
Lookback	96	336	96
MSE	0.355	0.300	0.272

336 timesteps reduced PatchTST's MSE by approximately 15.5%, indicating that forecasting performance for PatchTST was sensitive to input history length. Nevertheless, iTransformer continued to achieve the lowest forecasting error under the evaluated configurations.

8.3.2 Effects of Hidden State

A preliminary extension explored the introduction of a persistent hidden state into the synthetic data-generating process. While the resulting behaviour was qualitatively interesting, the experiment introduced multiple interacting factors simultaneously and was therefore less interpretable than the controlled λ and γ experiments.

Table 7 summarises forecasting performance for different levels of hidden-state persistence under the fully nonlinear configuration ($\lambda = 1$, $\gamma = 1$).

As hidden-state persistence decreased, forecasting performance deteriorated for all models. Although iTransformer consistently achieved the lowest forecasting error, its relative advantage diminished as state transitions became more frequent. One possible explanation is that lower persistence creates a more difficult latent-state inference problem, requiring either longer input histories or additional training before the benefits of more expressive architectures can be fully realised.These results should be treated as exploratory, as changing persistence simultaneously alters several properties of the data-generating process. Nevertheless, the experiments suggest that latent state dynamics may have a significant influence on forecasting performance. Future work could investigate synthetic datasets in which hidden states selectively activate different nonlinear interaction functions, providing a more systematic framework for studying regime changes and latent structure in multivariate forecasting tasks.

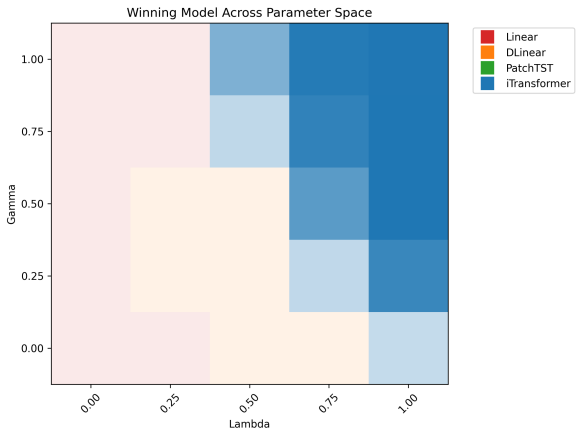


Figure 3: Winning model across the synthetic dataset parameter sweep. Cell colour indicates the model achieving the lowest MSE, while colour intensity indicates the margin of victory relative to the second-best model.

Table 7: Forecasting performance as a function of hidden-state persistence.

Persistence	Linear	DLinear	PatchTST	iTrans.
0.90	1.018	1.018	1.028	0.995
0.95	0.998	0.998	1.007	0.986
0.99	0.830	0.830	0.839	0.766
1.00	0.347	0.348	0.355	0.272

9 Discussion

9.1 Interpretation of Results

The published iTransformer results reported by Liu et al. were successfully reproduced in this study. All results typically differed from the published values by less than 5%. Minor changes in model ranking were observed for some datasets, particularly for PatchTST, which exhibited the largest deviations from the published results. However, the PatchTST results were obtained using the official PatchTST implementation rather than code provided by Liu et al., and the precise hyperparameter configuration used in the original study is not fully specified in the original paper. Accordingly, some variation in forecasting performance may be expected.

The evaluation of iTransformer on the new real-world datasets showed that, although the model achieved reasonable forecasting accuracy, it did not outperform the simpler linear forecasting approaches or PatchTST. It should be noted that the datasets were not selected *a priori* on the basis of forecast difficulty, non-

linearity, or known weaknesses of linear models. Subsequent analysis showed that the reconstructed Beijing dataset exhibited substantial linear predictability, with many variables achieving very high autoregressive R^2 values. This provides a plausible explanation for the strong performance of linear forecasting methods on this dataset. In contrast, the Space Weather dataset exhibited a much wider range of predictability, with several variables displaying considerably weaker linear structure. Nevertheless, the iTransformer model did not demonstrate a clear or consistent advantage. These findings suggest that forecasting performance is influenced by factors beyond linear predictability alone.

The results of the synthetic experiments provide additional insight into these observations. In the presence of only moderate nonlinearities, iTransformer did not outperform the linear baselines. It was only when strong nonlinear interactions were introduced that iTransformer consistently outperformed all three baseline models. These findings suggest that iTransformer may be more effective in forecasting problems containing strong nonlinear relationships between variates. Interestingly, PatchTST exhibited the opposite pattern, performing strongly on the real-world datasets but showing comparatively weaker performance in the synthetic experiments. This may indicate that the synthetic data generation process does not fully capture all of the characteristics present in real-world forecasting problems.

However, the present experiments do not establish nonlinearity as the sole determinant of performance. Alternative model configurations, including larger architectures, additional layers, different context lengths, or more extensive training data, may produce different results. A limitation of this study is that exhaustive architecture-specific hyperparameter optimisation was not performed for the new datasets. Although model configurations were chosen to provide a fair basis for comparison, they were not necessarily optimal for each architecture. Consequently, the reported results should be interpreted as comparative evaluations rather than estimates of the best achievable performance of each forecasting architecture.

9.2 LLM-as-Judge Evaluation

An LLM-as-judge framework was not used as part of the primary evaluation. Unlike many natural language processing tasks, time-series forecasting provides objective ground-truth observations together with well-established quantitative evaluation metrics. Consequently, forecasting performance was evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE), following the methodology of Liu et al.

LLM-based evaluation was not used for quantitative scoring or model ranking because forecasting accuracy can be measured directly against known future observations. In this context, objective statistical metrics provide transparent, reproducible and directly interpretable measures of predictive performance. The use of an LLM would introduce an additional layer of subjective interpretation without improving the reliability of the underlying accuracy assessment.

However, an LLM was used during dataset development as a qualitative sanity-checking tool. The model was prompted to review newly generated datasets for implausible patterns, potential generation artefacts, and inconsistencies with the intended data-generating processes. This review process identified a significant issue in the initial construction of the Beijing PM2.5 dataset, which prompted further manual investigation and subsequent correction. The LLM's role was limited to flagging potential concerns; all identified issues were independently verified using statistical analysis and inspection of the data-processing pipeline. No results reported in this thesis were derived from LLM-based evaluation.

10 Ethical Considerations

No major ethical concerns arose from either the original paper or the datasets used in this study. The original iTransformer evaluation was conducted using publicly available benchmark datasets that do not contain personally identifiable information, and the additional datasets used in this project consisted of publicly available environmental and scientific measurements rather than data relating to individuals.

The Australian Code for the Responsible Conduct of Research outlines eight principles that are hallmarks of responsible research conduct. The most significant ethical considerations pertaining to this study are honesty, rigour and transparency. The accompanying guidance on the management of research data emphasises the maintenance of accurate research records (National Health and Medical Research Council et al., 2018, 2019). These principles were particularly relevant because a substantial component of this project involved the reproduction of previously published results.

Consistent with the Australian Code for the Responsible Conduct of Research, experimental procedures, preprocessing steps, model configurations and dataset modifications were documented to enable independent verification of the reported findings. All modifications to existing datasets were recorded and reported. Code for parsing and processing the data, together with the experimental code used to generate the reported results, has been made available to allow independent reproduction and evaluation by future researchers. Experimental configurations were automatically recorded with each experiment run, ensuring that all reported results can be traced back to their original settings. Representative iTransformer configurations and results have been provided in Appendix C.

An exploratory analysis of the Space Weather dataset identified a potential issue affecting several derived variables. This issue was subsequently reported to the dataset maintainer. Consistent with the principles of honesty, transparency and accountability, potential data quality concerns were investigated, documented and disclosed rather than ignored.

An additional ethical responsibility concerns the reporting of negative or unexpected results. Several experiments showed that simple baseline models achieved performance comparable to, or better than, more sophisticated forecasting architectures. Consistent with the principles of honesty and transparency, these findings have been reported rather than omitted.

11 Conclusions and Further work

This study successfully reproduced the published iTransformer results and extended the evaluation to new real-world and synthetic forecasting datasets. Replicated iTransformer results on the datasets evaluated by Liu et al. typically differed from the published values by less than 5%, despite being obtained using different hardware and software environments.

Evaluation on the additional real-world datasets, Beijing PM2.5 and Space Weather, produced mixed results. iTransformer did not consistently outperform simpler baselines. These findings suggest that the advantages of iTransformer do not necessarily translate across all forecasting domains and may depend on the statistical characteristics of the underlying data.

Results from the synthetic experiments suggest that iTransformer performs best when strong nonlinear interactions exist between variates, achieving its best relative performance under the most nonlinear conditions. However, the differing behaviour of PatchTST between the synthetic and real-world datasets indicates that forecasting performance is influenced by characteristics not fully captured by the synthetic data generation process.

Future work could further investigate the conditions under which iTransformer outperforms alternative forecasting architectures. The advantages demonstrated by iTransformer on synthetic data did not consistently translate to the additional real-world datasets. More sophisticated synthetic data generators may help bridge the gap between the controlled synthetic experiments and the behaviour observed on real-world forecasting tasks. More extensive hyperparameter optimisation could also be conducted to better understand the sensitivity of different forecasting architectures to optimisation settings and input history length.

References

Monya Baker. 2016. 1,500 scientists lift the lid on reproducibility. *Nature*, 533(7604):452–454.

Andy Cockburn, Pierre Dragicevic, Lonni Besancon, and Carl Gutwin. 2020. Threats of a replication crisis in empirical computer science. *Communications of the ACM*, 63(8):70–79.

Guokun Lai, Wei-Cheng Chang, Yiming Yang, and Hanxiao Liu. 2018. Modeling long- and short-term temporal patterns with deep neural networks. In *International ACM SIGIR Conference on Research and Development in Information Retrieval*.

Minhao Liu, Ailing Zeng, Qi Wen, Zhijian Xu, Guokun Lai, Ling Zhao, and Chuang Ma. 2022. SCINet: Time series modeling and forecasting with sample convolution and interaction. In *Advances in Neural Information Processing Systems*.

Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. 2024. iTransformer: Inverted transformers are effective for time series forecasting. In *International Conference on Learning Representations*.

National Health and Medical Research Council, Australian Research Council, and Universities Australia. 2018. Australian code for the responsible conduct of research 2018. NHMRC Publication Reference R41.

National Health and Medical Research Council, Australian Research Council, and Universities Australia. 2019. Management of data and information in research: A guide supporting the australian code for the responsible conduct of research. NHMRC Publication Reference R41B.

Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2023. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*.

Julien Simon. 2026. Space Weather Indices (Kp, Ap, F10.7).

Chang Wei Tan, Christoph Bergmeir, Francois Petitjean, and Geoffrey I. Webb. 2020. Beijing PM2.5 dataset.

Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. 2021. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. In *Advances in Neural Information Processing Systems*.

Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. 2022. Are transformers effective for time series forecasting? *Preprint*, arXiv:2205.13504.

Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. 2021. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *AAAI Conference on Artificial Intelligence*.

Acknowledgement of LLM Use

Generative AI tools were used during the preparation of this report in accordance with university guidelines. Microsoft Copilot was used to assist with proofreading, improving grammar and writing style, providing feedback on the clarity and structure of draft sections. Copilot was also used to support qualitative review of the generated datasets and experimental findings. All references, dataset selections, interpretations of the literature, experimental design decisions, implementation work, and final report content were reviewed and verified by the author. The author takes full responsibility for all material presented in this report.

A Variate-Level Linear Predictability Results

Table 8: Summary statistics for variate-level linear predictability in the reconstructed Beijing PM2.5 dataset.

Statistic	R^2
Mean	0.816
Median	0.919
Minimum	0.073
25th Percentile	0.867
75th Percentile	0.990
Maximum	0.995

Table 9: Summary statistics for variate-level linear predictability in the Space Weather dataset.

Statistic	R^2
Mean	0.341
Median	0.224
Minimum	0.097
25th Percentile	0.119
75th Percentile	0.375
Maximum	1.000

B Complete Experimental Results for Synthetic Data

Table 10: Synthetic dataset results for persistence=1.

λ	γ	Linear		DLinear		PatchTST		iTransformer	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
0.00	0.00	0.995	0.795	0.996	0.796	1.008	0.801	1.007	0.801
0.00	0.25	0.995	0.795	0.996	0.796	1.008	0.801	1.007	0.801
0.00	0.50	0.995	0.795	0.996	0.796	1.008	0.801	1.007	0.801
0.00	0.75	0.995	0.795	0.996	0.796	1.008	0.801	1.007	0.801
0.00	1.00	0.995	0.795	0.996	0.796	1.008	0.801	1.007	0.801
0.25	0.00	0.996	0.796	0.996	0.796	1.008	0.801	1.008	0.801
0.25	0.25	0.996	0.796	0.996	0.796	1.008	0.801	1.008	0.801
0.25	0.50	0.997	0.797	0.997	0.797	1.009	0.801	1.009	0.801
0.25	0.75	0.997	0.797	0.997	0.797	1.009	0.802	1.009	0.802
0.25	1.00	0.997	0.797	0.997	0.797	1.010	0.802	1.008	0.802
0.50	0.00	0.997	0.796	0.997	0.796	1.009	0.801	1.008	0.801
0.50	0.25	0.997	0.797	0.997	0.797	1.009	0.802	1.007	0.801
0.50	0.50	0.995	0.797	0.995	0.797	1.008	0.801	0.998	0.798
0.50	0.75	0.988	0.794	0.988	0.794	1.001	0.799	0.974	0.788
0.50	1.00	0.971	0.787	0.971	0.787	0.984	0.793	0.921	0.767
0.75	0.00	0.997	0.797	0.997	0.797	1.009	0.802	1.006	0.800
0.75	0.25	0.989	0.794	0.989	0.794	1.002	0.799	0.975	0.788
0.75	0.50	0.943	0.776	0.944	0.776	0.958	0.782	0.859	0.740
0.75	0.75	0.811	0.720	0.811	0.720	0.825	0.726	0.675	0.655
0.75	1.00	0.663	0.649	0.663	0.649	0.677	0.657	0.531	0.579
1.00	0.00	0.991	0.795	0.991	0.795	1.004	0.800	0.978	0.789
1.00	0.25	0.870	0.745	0.871	0.745	0.885	0.751	0.741	0.686
1.00	0.50	0.528	0.576	0.528	0.577	0.539	0.582	0.413	0.508
1.00	0.75	0.380	0.486	0.380	0.486	0.389	0.492	0.296	0.429
1.00	1.00	0.347	0.464	0.348	0.464	0.355	0.470	0.272	0.411

Note: varying γ has no effect when $\lambda = 0$, as all cross-variate interactions are removed from the data-generating process. Under these conditions, the nonlinear interaction term makes no contribution to the generated time series.

C Reproducibility Information

The complete source code, dataset generation scripts, data loading pipelines and experiment launch scripts used throughout this project are available in the accompanying GitHub repositories:

- iTransformer: https://github.com/josephbelbin-mq/iTransformer
- PatchTST: https://github.com/josephbelbin-mq/PatchTST

Experimental configurations were automatically recorded during experimentation to support result traceability and reproducibility. The repositories contain the scripts used to generate datasets, launch experiments and evaluate forecasting performance. A machine-readable summary of recorded experimental results and configurations for the iTransformer runs is also provided:

- Experiment summary: https://github.com/josephbelbin-mq/iTransformer/blob/main/analysis/itransformer_results.csv

This summary file includes the dataset, model, recorded configuration, and evaluation metrics (MSE, MAE, RMSE, MAPE, and MSPE) for each experiment.